

1. Perceptron (artificial neuron):

A. perceptron is type of artificial neuron.

B. Input: $x_i \{i | 0, 1, \dots, n\}$ [bias input: $x_0 = 1$ (always)].

C. Weight: $w_i \{i | 0, 1, \dots, n\}$ [bias weight: w_0].

D. Computational Unit

weighted sum function: $z = \sum_0^n w_i \cdot x_i$

activation function: $f(z) = \begin{cases} -1, & z < 0 \\ 1, & z \geq 0 \end{cases}$

use "sign function" as an example
"sign function" is a common kind of activation function

E. Output: $y = f(z)$

F.

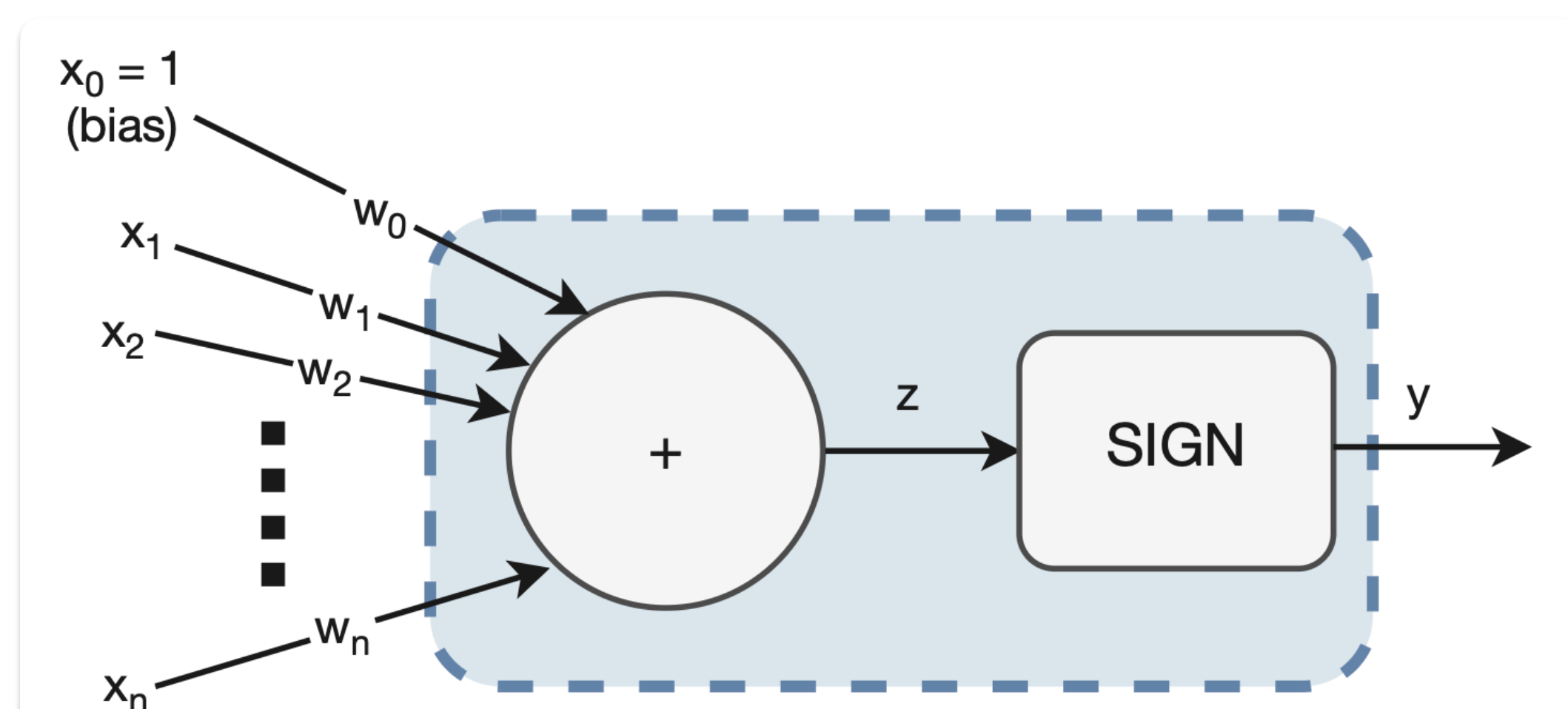


Figure 1-2 The perceptron

2. Neural Network (combined with few artificial neuron):

A. input & output of a perceptron can be "a range of real number."

B. combination of few perceptrons can implement more complex function than only single perceptron can do.

C. use "learning algorithm" & "observed example", weight of each perceptron can be adjust to design a more accurate neural network automatically.

D. the rebuilt neuron network tends to generalize that can make sense for "not-yet-observed example" and get correct outputs.